

HackEval AI

Multi-Step Prompt Workflow for Hackathon Idea Evaluation

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Assessment Type	Mini Assessment – Prompt Workflow Design

1. Project Objective

The objective of HackEval AI is to design a structured, multi-step prompt workflow that takes a hackathon problem statement and proposed solution as input, evaluates the idea from multiple perspectives, improves its weaknesses, and produces a concise, presentation-ready final proposal. The workflow demonstrates prompt chaining, where the output of one stage is deliberately passed as context to the next stage.

2. Problem Statement

Hackathon participants may have strong ideas but often struggle to determine whether their solution is sufficiently innovative, technically feasible, scalable, impactful, and clearly communicated. A single AI prompt may produce a useful answer, but it does not provide a systematic evaluation process. HackEval AI addresses this by dividing the reasoning process into six clearly defined stages.

3. Workflow Architecture

INPUT	PROBLEM UNDERSTANDING	SOLUTION ANALYSIS	INNOVATION EVALUATION	FEASIBILITY & IMPACT	OPTIMIZE IDEA	FINAL PITCH
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4. Workflow Steps and Expected Outputs

STEP	PURPOSE	EXPECTED OUTPUT
1. Problem Understanding	Break down the problem statement, users, challenges, root causes, constraints and success criteria.	Structured problem analysis containing the core need, users, constraints and expected outcome.
2. Solution Analysis	Examine the proposed solution against the problem and map features to user needs.	Solution assessment with features, strengths, user journey and identified gaps.
3. Innovation Evaluation	Assess uniqueness, differentiation and potential innovation of the proposed approach.	Innovation score (1–10) , USP, differentiation opportunities and improvement suggestions.
4. Feasibility & Impact	Evaluate technology, cost, complexity, scalability, adoption and real-world impact.	Feasibility and impact scorecard with risks and mitigation strategies.
5. Solution Optimization	Strengthen the original idea while preserving its fundamental purpose.	Improved solution concept with enhanced features, UX, scalability and technical approach.
6. Final Pitch Generator	Combine all validated findings into a concise hackathon presentation.	Final proposal containing USP, technology, impact, score and a 60-second pitch.

5. Detailed Prompt Design

Step 1 — Problem Understanding

Analyze the provided hackathon problem statement. Identify the core problem, target users or beneficiaries, current challenges, root causes, constraints, expected outcome, and key requirements that a successful solution should satisfy. Do not propose a solution yet. Return the analysis using clear headings and concise bullet points.

Step 2 — Proposed Solution Analysis

Using the problem analysis from the previous step, evaluate the proposed hackathon solution. Explain how it addresses the core problem, identify target users, main features, user journey, feature-to-problem mapping, strengths, and missing components. Do not completely redesign the solution at this stage; focus on evaluating the existing proposal.

Step 3 — Innovation & Uniqueness Evaluation

Evaluate the proposed solution for innovation and uniqueness. Identify its innovative aspects, features that are common or replaceable, unique value proposition, and opportunities for differentiation. Suggest three specific improvements that could increase novelty and provide an innovation score from 1–10 with justification. Avoid unsupported claims about competitors.

Step 4 — Feasibility & Impact Assessment

Evaluate the solution across technical feasibility, implementation complexity, cost feasibility, scalability, real-world impact, user adoption potential, and sustainability. Give each category a score from 1–10 with a concise justification. Calculate an overall score out of 10. Identify the three biggest implementation risks and one mitigation strategy for each.

Step 5 — Solution Optimization

Improve the proposed hackathon solution using the previous evaluations. Preserve the fundamental purpose while strengthening innovation, technical feasibility, scalability and user experience. Provide an improved concept, five improved features, new innovative elements, expected impact, and explain how the revised solution addresses earlier weaknesses.

Step 6 — Final Hackathon Pitch Generator

Using all previous analyses, generate the final hackathon-ready proposal. Include: project title, one-line problem statement, proposed solution, target users, key features, innovation/USP, technology approach, implementation approach, expected impact, scalability, risks and mitigations, overall evaluation score, and a persuasive 60-second pitch. Do not invent facts, statistics, partnerships or achievements.

6. Prompt Chaining & Information Flow

The workflow is designed so that each stage builds on verified context from earlier stages. Step 1 creates the problem analysis; Step 2 consumes that analysis to evaluate the solution; Steps 3 and 4 use the solution analysis to score innovation, feasibility and impact; Step 5 uses those findings to improve the concept; and Step 6 consolidates the optimized concept into the final pitch. This makes the workflow reproducible, traceable and more reliable than a single broad prompt.

7. Demonstration Example

Input Problem: Farmers often struggle to identify crop diseases at an early stage. Limited access to agricultural experts and delayed diagnosis can result in crop losses. Develop an AI-powered solution that helps farmers detect crop diseases early and provides actionable recommendations.

Proposed Solution: SmartFarm AI is a mobile-based AI assistant that analyzes crop images using computer vision to identify diseases and provides farmers with recommendations for treatment and prevention.

Expected Final Result: After the six stages, the system should produce an evaluated and improved SmartFarm AI concept with a clear USP, technology approach, impact assessment, feasibility score, risk mitigation plan and a concise judge-facing pitch.

8. Expected Learning Outcome

This workflow demonstrates how prompt engineering can be used to decompose a complex task into manageable AI interactions. It highlights sequential prompting, contextual reuse, structured outputs, evaluation criteria, iterative improvement and final content generation—all without requiring software development or code.

9. Conclusion

HackEval AI transforms an initial hackathon concept into a structured, evaluated and pitch-ready proposal through six connected prompt stages. By separating problem understanding, solution analysis, innovation assessment, feasibility evaluation, optimization and final communication, the workflow provides a clear and professional approach to AI-assisted hackathon preparation.